

Iterative Self-Refinement for Multi-Camera Calibration via Learned Pose Conditioning

Abstract

Multi-camera extrinsic calibration fails in wide-baseline deployments where dynamic, visually repetitive content dominates the scene and static geometry is degenerate. Feed-forward geometry models such as Pi3X can estimate poses in these settings, but their accuracy saturates under standard post-processing. We show that feeding a model’s own predictions back as pose conditioning creates an iterative refinement loop with a surprising asymmetry: rotation contracts toward the correct solution on $SO(3)$ (contraction rate $\lambda \approx 0.8$), while translation diverges due to $Sim(3)$ scale ambiguity. To exploit this, we propose *decoupled hybrid fusion*—sourcing rotation edges from conditioned inference and translation edges from unconditioned inference within a unified pose graph. Iterating this fusion achieves 27% rotation improvement and 36% relative pose improvement over the raw model output on a challenging 8-camera feedlot dataset, at only $3\times$ the compute cost and with no external calibration data required. Through layer ablation, we trace the conditioning benefit to the *encode path*—ray embeddings and scale normalisation derived from approximate poses reduce feature ambiguity before decoding—while the decode-path pose injection blocks contribute nothing. This identifies self-refinement as block coordinate descent on a joint pose-feature objective and explains both the rotation contraction and the translation divergence mechanistically.

1 Introduction

Extrinsic calibration of wide-baseline multi-camera systems is a prerequisite for 3D reconstruction, tracking, and scene understanding. Classical approaches rely on either physical calibration targets [19] or structure-from-motion pipelines [9] that extract and match distinctive static features across views. Both strategies fail when cameras are spaced far apart with minimal field-of-view overlap, when the dominant visual content is dynamic and repetitive, and when the static scene geometry is degenerate. These three conditions co-occur in a growing class of deployments—livestock monitoring, wildlife observation, sports analytics, and large-area surveillance—where cameras observe moving subjects of near-identical appearance over geometrically unstructured backgrounds.

We study this problem in the concrete setting of eight wide-baseline cameras monitoring a cattle feedlot. The scene exhibits all three failure modes simultaneously: cameras are separated by 5–10 m with narrow overlap, each frame is dominated by moving cattle of near-identical black appearance, and the static pen is an approximately planar surface with few distinctive landmarks. Classical SfM pipelines produce no usable output in this setting.

Recent feed-forward geometry models [17, 14, 15] bypass explicit feature correspondence by learning to infer multi-view geometry directly from images. Pi3X [17], a permutation-equivariant model that processes all N views jointly, produces camera poses with $\sim 4.4^\circ$ mean rotation error from raw images alone—succeeding where classical methods fail entirely. However, standard geometric post-processing (robust averaging, pose graph optimization) yields no further rotation improvement,

because the residual error is systematic bias from weak geometric signal rather than statistical noise amenable to averaging.

We observe that Pi3X accepts optional pose conditioning: feeding approximate camera poses alongside images improves its predictions. This raises the natural question of whether the model can refine *its own output* through iterative self-conditioning—analogous to fixed-point iteration on the pose manifold. We discover a fundamental asymmetry in this process: **self-conditioning contracts rotation toward the correct solution but causes translation to diverge**. The divergence arises because Pi3X’s outputs are defined only up to $\text{Sim}(3)$ —rotation is scale-invariant and thus stable under re-injection, while translation accumulates compounding scale bias when fed back as $\text{SE}(3)$ conditioning.

This asymmetry motivates two questions that we address in this paper: *Under what conditions does self-conditioning converge on the rotation manifold?* And: *How can the rotation–translation asymmetry be exploited to improve calibration without degrading either component?*

Our contributions are as follows:

1. We provide the first empirical characterisation of self-conditioning dynamics in learned geometry models, showing that conditioned inference acts as a contraction mapping on $\text{SO}(3)$ with measurable rate $\lambda \approx 0.8$, while the translation component does not contract due to $\text{Sim}(3)$ scale ambiguity.
2. We identify the mechanistic pathway: conditioning operates through the *encode path*—ray embeddings derived from approximate poses reduce feature ambiguity before decoding. Layer ablation reveals that the decode-path PoseInjectBlocks (projective rotary position encoding) contribute nothing when zeroed. This identifies self-refinement as block coordinate descent on a joint pose-feature objective.
3. We propose *decoupled hybrid fusion*, a method that exploits the rotation–translation asymmetry by sourcing rotation edges from conditioned inference and translation edges from unconditioned inference within a unified pose graph. Iterating this fusion achieves 27% rotation improvement and 36% relative pose improvement over the baseline at only $3\times$ compute cost, without any external calibration data.
4. We derive the method’s accuracy floor from the measured contraction rate and validate the prediction experimentally, establishing principled criteria for when self-refinement helps and when it saturates.

2 Related Work

Learned multi-view geometry. Feed-forward models for multi-view 3D reconstruction have advanced rapidly. DUST3R [15] introduced pairwise pointmap regression; MAST3R [6] extended it with local feature matching. Pi3X [17] and VGGT [14] process all N views jointly via permutation-equivariant architectures, achieving stronger global consistency. These models output camera poses and dense geometry from images alone, but their accuracy on wide-baseline, feature-adversarial scenes remains limited.

Multi-camera calibration. Classical extrinsic calibration uses physical targets (checkerboards [19]), vanishing points [3], or structure-from-motion pipelines [9]. Target-free methods assume sufficient static texture for feature matching across views—an assumption violated in dynamic, textureless environments. Our setting falls outside the scope of these methods.

Pose graph optimization. Rotation averaging [5, 2] and translation averaging [8] are classical tools for aggregating noisy pairwise relative poses into a globally consistent estimate. We adopt these as our post-processing backbone (Stage 1) and as the fusion mechanism for combining heterogeneous edge sources (Stages 2–3).

Test-time refinement. Test-time training (TTT) [11] and test-time adaptation adapt model predictions at inference time using self-supervised objectives. In geometry estimation, self-supervised depth refinement [4] fine-tunes networks on test sequences using photometric consistency. Our approach differs in that we do *not* modify model weights—instead, we exploit the model’s conditioning interface to iteratively refine predictions, treating the pretrained model as a fixed map whose encode-path features improve under better pose priors.

Iterative pose refinement. Classical approaches such as ICP [1] and PnP+RANSAC refinement iterate between correspondence estimation and pose optimization. Bundle adjustment [12] jointly optimises poses and 3D structure. Our method is conceptually similar—iterating between feature computation (given poses) and pose estimation (given features)—sharing the structure of block coordinate descent [16] rather than explicit geometric optimization.

Denoising and diffusion on manifolds. Score-based diffusion models [10] learn to denoise data via iterative refinement from noise toward the data manifold. Recent work has extended diffusion to SE(3) [18] for molecular and pose generation. Our iterative self-conditioning superficially resembles this structure. However, we show empirically that the denoising interpretation is misleading: the benefit arises from encode-path feature improvement (ray embeddings derived from approximate poses), not from iterative noise removal. Monte Carlo perturbation provides no benefit (Section 7.6), confirming that the mechanism is deterministic feature enrichment rather than stochastic denoising.

3 Foundation Geometry Model Benchmark

To select the best backbone for our calibration pipeline, we benchmark seven recent foundation geometry models on our dataset. All models receive the same 8 synchronised camera images per timestep and must output camera poses without any prior information.

3.1 Experimental Protocol

We sample 91 timesteps at regular intervals from synchronised video (2688×1520 resolution, resized per model requirements). For each model, we:

1. Run inference on all timesteps (8 cameras per timestep)
2. Aggregate per-timestep poses via quaternion averaging (rotations) and median (translations)
3. Apply Sim(3) Procrustes alignment to ground truth (with reflection check)
4. Compute ATE, ARE, RPE, and co-located pair distances

Ground truth poses are obtained from COLMAP applied to a static scene capture (no cat-tle present). Two camera pairs (C2–C3 and C5–C6) share physical mounting poles, providing a structural validation metric (ground truth distance ≈ 0).

Table 1: Foundation geometry model benchmark on our wide-baseline, feature-adversarial dataset. Models are grouped by type (N-view joint vs. pairwise with post-hoc alignment) and ranked by ATE within each group. “Cond.” indicates whether the model accepts pose/intrinsic priors as conditioning input—a prerequisite for our self-refinement method. All metrics use Sim(3)-aligned predictions; C2C3 measures co-located camera consistency. Pairwise models use only 20 timesteps due to computational cost.

Model	Venue	Type	Cond.	ATE↓	ARE↓	RPE _R ↓	RPE _t ↓	C2C3↓	Time (s)
<i>N-view (joint)</i>									
Pi3X	CVPR’25	N-view	✓	0.177	4.37°	6.26°	0.39	0.088	106
Pi3	CVPR’25	N-view	—	0.242	4.08°	5.98°	0.37	0.000	121
MapAnything	CVPR’25	N-view	✓	0.627	9.17°	14.19°	0.93	0.313	86
VGGT	CVPR’25	N-view	—	1.248	10.48°	12.82°	1.55	2.011	75
NoPo4D	arXiv’25	N-view	—	2.556	62.06°	77.01°	3.17	0.627	214
Fast3R	CVPR’25	N-view	—	2.743	96.64°	102.88°	4.36	3.183	460
<i>Pairwise (post-hoc alignment)</i>									
Pow3R	CVPR’25	Pairwise	✓	1.835	180°	28.53°	7.24	0.218	497
MonST3R	ICLR’25	Pairwise	—	2.607	47.24°	23.87°	3.56	0.111	1492

3.2 Results

3.3 Analysis

The results reveal a clear performance gap between architectures. Pi3X and Pi3 (its smaller variant) both achieve sub-5° rotation error, with Pi3 achieving slightly better ARE (4.08° vs. 4.37°) but worse ATE (0.242 vs. 0.177). MapAnything and VGGT form a second cluster at ~10° rotation error: functional but substantially less accurate. The remaining models (NoPo4D, Fast3R, MonST3R, Pow3R) all exceed 20° rotation error and effectively fail on this dataset.

Two structural patterns emerge from these results. First, N-view joint architectures (Pi3X, MapAnything, VGGT) consistently outperform pairwise models, confirming that joint multi-view reasoning is essential when inter-camera overlap is minimal—pairwise models cannot propagate geometric constraints across views that share no visual content. Second, the co-location metric reveals differences in structural understanding that are masked by aggregate error: Pi3X correctly infers that cameras C2 and C3 share a mounting pole (distance 0.088), while VGGT (2.011) and Fast3R (3.183) place them far apart.

Of the eight models tested, only three support pose conditioning (Table 1, “Cond.” column): Pi3X, MapAnything, and Pow3R. Pi3 (base), despite competitive rotation accuracy, lacks a conditioning interface and thus cannot benefit from self-refinement. Pi3X combines the best overall accuracy with conditioning support, making it the natural backbone. Note that Pi3 (base) does not support conditioning despite being from the same model family—the conditioning mechanism was introduced in the Pi3X extension. The remainder of this paper develops methods to push beyond Pi3X’s ~4.4° accuracy ceiling, which resists further improvement through standard geometric post-processing alone.

4 Method

Our method operates in three stages. Stage 1 applies standard geometric post-processing to establish a baseline estimate. Stages 2–3 constitute our contribution: self-conditioned iterative refinement with decoupled rotation-translation fusion.

4.1 Preliminaries: Geometric Post-Processing (Stage 1)

Given $N_t = 91$ timesteps of N -camera images, we run Pi3X to obtain per-timestep poses $\{T_k^c\}_{k=1}^{N_t}$ for each camera $c \in \{1, \dots, N\}$, where each $T_k^c \in \text{SE}(3)$ is defined up to a per-timestep $\text{Sim}(3)$ ambiguity (i.e., each timestep’s output coordinate frame is arbitrary in position, orientation, and scale). We aggregate these into a single estimate via three standard steps.

Gauge fixing. Since each timestep uses an arbitrarily chosen global frame, we align all timesteps to a reference via $\text{Sim}(3)$ Procrustes [13]:

$$\{s^*, R^*, \mathbf{t}^*\} = \arg \min_{s, R, \mathbf{t}} \sum_{c=1}^N \|sR\mathbf{p}_k^c + \mathbf{t} - \mathbf{p}_{\text{ref}}^c\|^2 \quad (1)$$

where $\mathbf{p}_k^c \in \mathbb{R}^3$ denotes the position of camera c at timestep k , and $s \in \mathbb{R}^+$, $R \in \text{SO}(3)$, $\mathbf{t} \in \mathbb{R}^3$ parametrise the $\text{Sim}(3)$ transform.

Outlier rejection. For each camera, we project its N_t estimated positions onto the dominant 2D plane (via PCA on camera centres) and reject outliers whose in-plane distance exceeds $3 \times$ the Median Absolute Deviation (MAD). Inlier rotations are aggregated via the Fréchet mean on $\text{SO}(3)$ [7]—the rotation minimising the sum of squared geodesic distances to all inlier samples.

Pose graph optimization. We extract pairwise relative poses $T_{ij} = (T^i)^{-1}T^j$ from inlier timesteps, computing the Fréchet mean rotation and mean translation for each camera pair (i, j) . These relative poses serve as edges in a pose graph, weighted by the inverse variance of the per-timestep estimates (edges with higher consistency receive more weight). We then optimise globally via L_1 Weiszfeld rotation averaging on $\text{SO}(3)$ [5] followed by translation averaging with hard co-location constraints for cameras sharing a physical mounting pole.

Residual error analysis. This pipeline reduces ATE by 10% ($0.177 \rightarrow 0.161$) but leaves ARE unchanged ($4.37^\circ \rightarrow 4.39^\circ$). The persistence of rotation error under robust averaging over 91 independent timesteps indicates that the residual is systematic model bias rather than sample noise. Since the scene provides weak geometric signal (approximately planar static structure, dynamic repetitive content), further classical aggregation cannot improve upon this estimate. This motivates the use of Pi3X’s conditioning interface to inject prior geometric information, as developed in the following sections.

4.2 Self-Conditioned Inference (Stage 2)

Pi3X supports multimodal conditioning: given approximate poses $\hat{T}^c$ and camera intrinsic matrices $\mathbf{\Pi}^c \in \mathbb{R}^{3 \times 3}$, it refines its predictions via two encode-path mechanisms: (1) ray embeddings that convert pixel coordinates to 3D directions using intrinsics and poses, added to image features before decoding; and (2) scale normalisation that uses pose translations to set proper depth scale. We

verify empirically that the decode-path PoseInjectBlocks (PRoPE attention injection) contribute nothing when ablated (Section 7.1). We condition on the Stage 1 output:

$$\{T_k^{c,(1)}\} = f_\theta(\text{imgs}_k \mid \hat{T}^c, \mathbf{\Pi}^c), \quad k = 1, \dots, N_t \quad (2)$$

where f_θ denotes Pi3X and $\hat{T}^c$ is the Stage 1 estimate converted to the model’s input camera ordering.

Applying the same geometric post-processing to $\{T_k^{c,(1)}\}$ yields a refined estimate with ARE reduced from 4.39° to 3.49° (20% improvement) but ATE increased from 0.161 to 0.166 (3% degradation). Iterating this process naïvely amplifies both effects: rotation improves further (to $\sim 3.3^\circ$) while translation degrades monotonically (to 0.195 ATE after 5 iterations). Section 5 provides a formal explanation: the rotation component contracts toward a fixed point on $\text{SO}(3)$, while translations diverge due to $\text{Sim}(3)$ scale mismatch in the conditioning pathway.

4.3 Decoupled Hybrid Fusion (Stage 3)

Since conditioned inference improves rotation but degrades translation, we decouple them in the pose graph by sourcing edges from different inference runs. Let R_{ij} and $\mathbf{t}_{ij}$ denote the relative rotation and translation from camera i to camera j , computed as the mean over inlier timesteps. We construct fused edges:

$$\tilde{T}_{ij}^{\text{fused}} = \begin{pmatrix} R_{ij}^{\text{cond}} & \mathbf{t}_{ij}^{\text{uncond}} \\ \mathbf{0}^\top & 1 \end{pmatrix} \quad (3)$$

where R_{ij}^{cond} is the mean relative rotation extracted from conditioned inference (Stage 2) and $\mathbf{t}_{ij}^{\text{uncond}}$ is the mean relative translation from the original unconditioned run (Stage 1). Edge weights w_{ij} (inverse variance of the per-timestep relative pose estimates) are taken as $w_{ij} = \max(w_{ij}^{\text{cond}}, w_{ij}^{\text{uncond}})$.

The initial pose for graph optimization similarly uses conditioned rotations with unconditioned translations:

$$T_{\text{init}}^c = \begin{pmatrix} R_{\text{cond}}^c & \mathbf{t}_{\text{uncond}}^c \\ \mathbf{0}^\top & 1 \end{pmatrix} \quad (4)$$

This yields $\text{ARE} = 3.45^\circ$ (rotation improved) with $\text{ATE} = 0.159$ (translation preserved)—capturing the benefits of both sources.

4.4 Iterated Hybrid Refinement

Since a single conditioned pass reduces rotation error by 20%, a natural question is whether iterating the process yields further gains. We apply hybrid fusion iteratively: at each step, the optimised rotation estimate from the previous iteration serves as the new conditioning input. This implements block coordinate descent on an implicit joint pose-feature objective: each iteration updates features (given poses) via the encode-path ray embeddings, then updates poses (given improved features) via the decoder.

Design choices.

- **Translation anchoring:** Translations are *never* updated from conditioned outputs—only from the original unconditioned baseline. This is the key invariant that prevents $\text{Sim}(3)$ drift.
- **Early stopping:** Terminate when ARE improvement between iterations falls below 0.1° , or after $M = 2$ iterations (empirically sufficient; see Section 7).

Algorithm 1 Iterated Hybrid Refinement

Require: Pipeline estimate $\hat{T}^c$, images, intrinsics $\mathbf{\Pi}^c$

Require: Iterations: M

```
1:  $T_{\text{curr}} \leftarrow \hat{T}^c$ 
2: Compute unconditioned edges  $\{\mathbf{t}_{ij}^{\text{uncond}}, w_{ij}^t\}$  {fixed throughout}
3: for  $m = 1$  to  $M$  do
4:    $\{T_k^{c,(m)}\} \leftarrow f_{\theta}(\text{imgs} \mid T_{\text{curr}}, \mathbf{\Pi}^c)$  for all  $k$ 
5:   Extract rotation edges  $\{R_{ij}^{\text{cond}}, w_{ij}^R\}$  from pipeline-processed conditioned poses
6:   Fuse:  $\tilde{T}_{ij} = (R_{ij}^{\text{cond}}, \mathbf{t}_{ij}^{\text{uncond}})$ 
7:    $T_{\text{opt}} \leftarrow \text{PoseGraph}(\tilde{T}_{\text{init}}, \tilde{T}_{ij}, w_{ij})$ 
8:    $T_{\text{curr}} \leftarrow (R_{\text{opt}}, \mathbf{t}_{\text{uncond}})$  {anchor translations}
9: end for
10: return  $T_{\text{opt}}$ 
```

Stochastic extension. One might expect that perturbing conditioning poses and averaging across multiple samples (Monte Carlo integration) could further reduce error. We investigate this extension in Section 7.6 and find it provides no benefit: deterministic iteration ($S = 1$) is optimal, consistent with the block coordinate descent interpretation where the mechanism is deterministic feature enrichment rather than stochastic denoising.

5 Theoretical Analysis

We provide a formal interpretation of the self-conditioning dynamics observed in Section 4, explaining why rotation converges while translation diverges, and identifying the mechanistic pathway through which conditioning operates.

5.1 Contraction Mapping on $\text{SO}(3)$

Let $f : \text{SO}(3)^N \rightarrow \text{SO}(3)^N$ denote the composite mapping from conditioning rotations to Pi3X’s output rotations after pipeline processing (gauge fix, outlier rejection, Fréchet mean). We define the geodesic distance on $\text{SO}(3)$ as:

$$d_{\text{SO}(3)}(R_1, R_2) = \|\log(R_1^\top R_2)\|_F \quad (5)$$

where $\log : \text{SO}(3) \rightarrow \mathfrak{so}(3)$ is the matrix logarithm. This equals the angle of the relative rotation $R_1^\top R_2$.

If f is a contraction mapping, i.e., there exists $\lambda < 1$ such that for all R in a neighbourhood of the true rotations R^* :

$$d_{\text{SO}(3)}(f(R), R^*) \leq \lambda \cdot d_{\text{SO}(3)}(R, R^*) \quad (6)$$

then by the Banach fixed-point theorem, the iteration $R^{(n+1)} = f(R^{(n)})$ converges to a unique fixed point R^∞ satisfying $f(R^\infty) = R^\infty$.

Empirical measurement. We measure the contraction rate by feeding GT poses corrupted with known noise levels and comparing input error to output error (Section 7, Table 5):

- Input error $4.4^\circ \rightarrow$ output error 3.5° ($\lambda \approx 0.80$)

- Input error $5.0^\circ \rightarrow$ output error 3.3° ($\lambda \approx 0.67$)
- Input error $2.0^\circ \rightarrow$ output error 1.6° ($\lambda \approx 0.80$)

The contraction rate is approximately $\lambda \approx 0.8$ in the 2–5° regime. This predicts convergence to a fixed point at:

$$\text{ARE}^* = \frac{\lambda}{1 - \lambda} \cdot \epsilon_{\text{irr}} \approx 4 \cdot \epsilon_{\text{irr}} \quad (7)$$

where ϵ_{irr} is the model’s irreducible error (the residual even with perfect conditioning). With $\text{ARE}^* \approx 3.1^\circ$ observed experimentally, this gives $\epsilon_{\text{irr}} \approx 0.8^\circ$, consistent with the 1.2° achieved under perfect GT conditioning (which includes additional pipeline processing noise).

Basin of attraction. The contraction property holds only within a finite neighbourhood of R^* . Our noise robustness experiments (Section 7) show that conditioning with $> 10^\circ$ error causes catastrophic failure (output error jumps to 180°). This defines a basin of attraction of approximately $\pm 10^\circ$ around the true solution—our initial estimate (4.4° error) lies safely within it.

5.2 Failure of Contraction for Translation

Pi3X outputs poses in $\text{Sim}(3)$, not $\text{SE}(3)$. The translation component $\mathbf{t}^c$ is meaningful only relative to the per-run scale s . When fed back as $\text{SE}(3)$ conditioning, a scale mismatch accumulates:

$$\mathbf{t}^{(n+1)} = s^{(n)} \cdot g(\mathbf{t}^{(n)}, R^{(n)}) + \beta \quad (8)$$

where $s^{(n)} \neq 1$ in general and β represents systematic bias. Since the scale factor $s^{(n)}$ varies per iteration and is not corrected, this does not satisfy contraction:

$$\|\mathbf{t}^{(n+1)} - \mathbf{t}^*\| \not\leq \lambda \|\mathbf{t}^{(n)} - \mathbf{t}^*\| \quad (9)$$

Mechanistic explanation. The encode path uses conditioning translations to normalise depth scale: $\mathbf{t}' = \mathbf{t} / \text{median}(\|\mathbf{t}\|)$ before constructing ray embeddings. Since Pi3X outputs in $\text{Sim}(3)$, the scale of the output translation is arbitrary relative to the conditioning scale. Feeding output translations back as input creates a compounding scale mismatch—each iteration’s normalisation references a different implicit scale, causing monotonic drift. Rotation is immune because it is scale-invariant: the rotation subgroup of $\text{Sim}(3)$ equals that of $\text{SE}(3)$.

This analysis motivates our hybrid fusion strategy: rotation contracts and should be iterated; translation does not and should be anchored to the unconditioned estimate.

5.3 Block Coordinate Descent Interpretation

Self-conditioning admits a natural interpretation as block coordinate descent (BCD) [16] on an implicit joint objective over poses and features. Let $\mathcal{F}(R, \phi)$ denote the implicit objective that Pi3X minimises during pretraining, where $R \in \text{SO}(3)^N$ are camera rotations and ϕ are internal feature representations. Each iteration alternates between two steps:

1. **Feature update** (given poses): The encode path constructs ray embeddings $\mathbf{r}_i = \text{lift}(\Pi^c) \cdot [R^{(n)} \mid \mathbf{t}]$ and normalises depth scale using $\|\mathbf{t}\|$, producing improved features $\phi^{(n+1)} = h(R^{(n)}, \text{imgs})$.
2. **Pose update** (given features): The decoder estimates poses from the enriched features: $R^{(n+1)} = g(\phi^{(n+1)})$.

The composite map $f = g \circ h$ maps conditioning poses to output poses. Contraction ($\lambda < 1$) arises because ray embeddings provide angular priors that collapse directional ambiguity in the feature space—each iteration’s features are strictly more informative than the previous iteration’s, yielding strictly better pose estimates until the fixed point.

Why rotation contracts but translation diverges. Ray embeddings encode pixel-to-3D *direction*, which depends on rotation but is scale-invariant. Better rotation priors yield better directional features, which yield better rotation estimates—a self-reinforcing loop with diminishing returns (contraction). Translation enters only through scale normalisation, where the Sim(3) ambiguity prevents convergence (Section 4.2). This explains the $\lambda \approx 0.8$ contraction rate as the composite Lipschitz constant of the encode-decode loop restricted to SO(3).

Consistency with ablation evidence. The BCD interpretation predicts that the active mechanism is the encode path (where poses $\rightarrow$ features), not the decode-path PoseInjectBlocks (where poses modulate attention). Our layer ablation (Section 7.1) confirms this: zeroing all five PoseInjectBlocks produces identical accuracy to leaving them active (ARE 3.84° vs 3.85°), while removing encode-path conditioning entirely (via identity poses) eliminates all improvement. The decode-path blocks are vestigial for this task.

5.4 Posterior Unimodality Under Self-Conditioning

One might expect that perturbing the conditioning signal and averaging—Monte Carlo integration over the model’s implicit posterior $p(R^* \mid R + \epsilon, \text{imgs})$ —would provide additional variance reduction. If the posterior has non-trivial spread, averaging S samples should reduce error as $O(1/\sqrt{S})$.

However, the BCD interpretation explains why this fails: since the mechanism is deterministic feature enrichment (ray embeddings from poses), not stochastic sampling, all perturbations within the basin of attraction produce the same enriched features and converge to the same output. The posterior is effectively a point mass (angular standard deviation $< 0.02^\circ$ across samples), leaving no variance for MC averaging to reduce. Section 7.6 confirms empirically that $S = 1$ is optimal.

6 Results

6.1 Experimental Setup

Dataset. We evaluate on 8 static cameras (2688×1520 resolution) deployed in a cattle feedlot. We sample $N_t = 91$ timesteps at regular intervals from synchronised video. Ground truth poses are obtained from COLMAP applied to a static capture of the same scene (without cattle).

Evaluation metrics. All methods are evaluated after Sim(3) Procrustes alignment to ground truth (with reflection check):

- **ATE:** Absolute Translation Error (RMSE of camera position errors)
- **ARE:** Absolute Rotation Error (mean geodesic distance to GT rotations, in degrees)
- **RPE rot / trans:** Relative Pose Error averaged over all $\binom{8}{2} = 28$ camera pairs
- **C2C3:** Distance between co-located cameras C2 and C3 (GT ≈ 0 ; structural consistency test)

Note that ATE and RPE trans are in arbitrary units (scene scale is determined by the Sim(3) alignment, not metric ground truth).

Table 2: Progressive improvement from raw Pi3X through our full method. Cost is relative to a single unconditioned inference pass (~ 100 s on an A100). Row 3 shows naïve self-conditioning (rotation improves but translation degrades). Row 4 introduces hybrid fusion (translation preserved). Row 5 iterates the hybrid method twice ($M=2$).

Method	ATE↓	ARE↓	RPE _R ↓	RPE _t ↓	C2C3↓	Cost
1. Raw Pi3X (quat average)	0.177	4.37°	6.26°	0.395	0.088	1×
2. + Pipeline (Stage 1)	0.161	4.39°	6.26°	0.375	0.000	1×
3. + Self-Conditioning (Stage 2)	0.166	3.49°	4.93°	0.384	0.000	2×
4. + Hybrid Fusion (Stage 3)	0.159	3.45°	4.93°	0.322	0.000	2×
5. + Iterated Hybrid ($M=2$)	0.151	3.19°	3.84°	0.254	0.000	3×
<i>Oracle (GT pose conditioning)</i>	<i>0.097</i>	<i>1.22°</i>	—	—	—	1×

6.2 Comprehensive Comparison

Table 2 presents the progressive improvement from raw Pi3X output through our full method.

6.3 Analysis

The progressive results in Table 2 reveal several findings.

Rotation error is not reducible by classical aggregation. The pipeline (row 2) reduces ATE by 10% through pose graph optimization with co-location constraints, but provides no rotation improvement ($4.37^\circ \rightarrow 4.39^\circ$). This confirms that Pi3X’s rotation error is systematic model bias rather than statistical noise: robust averaging over 91 timesteps has already extracted all available information from the raw outputs.

Self-conditioning exhibits rotation–translation asymmetry. Conditioning on the pipeline estimate (row 3) reduces ARE by 20% but increases ATE by 3%. This asymmetry—the central observation motivating our method—is consistent across multiple runs and noise configurations (Section 7).

Hybrid fusion resolves the asymmetry. Decoupling rotation and translation channels (row 4) achieves 3.45° ARE (capturing the conditioned rotation improvement) while maintaining ATE at 0.159 (preserving the unconditioned translation quality). The computational overhead is minimal: one additional conditioned inference pass ($2\times$ total cost).

Iteration further improves rotation at minimal cost. The full method (row 5) applies hybrid fusion twice ($M = 2$), reducing ARE by an additional 8% over single-step hybrid fusion ($3.45^\circ \rightarrow 3.19^\circ$). Each iteration improves the encode-path ray embeddings, which in turn yield better features and poses—block coordinate descent on the joint pose-feature objective (Section 5.3). Monte Carlo perturbation provides no additional benefit (Section 7.6), consistent with the deterministic nature of the mechanism. The $3\times$ total cost makes iterated hybrid fusion practical even for interactive applications.

Aggregate improvement. Comparing the raw model output (row 1) to the full method (row 5): ARE improves by 27% ($4.37^\circ \rightarrow 3.19^\circ$), RPE rotation by 39%, RPE translation by 36%, and ATE by 15%. The co-location constraint (C2C3) is satisfied exactly under pose graph optimization (rows 2–5).

Oracle upper bound. The final row reports Pi3X conditioned on ground truth poses, representing the best achievable accuracy given perfect conditioning. This oracle achieves $\text{ARE} = 1.22^\circ$ and $\text{ATE} = 0.097$ —reductions of 72% and 40% over the unconditioned baseline. The gap between our self-refined result (3.19°) and the oracle (1.22°) quantifies the remaining headroom attributable to conditioning quality. The monotonic improvement from unconditioned (4.39°) through self-conditioned (3.19°) to oracle (1.22°) confirms that the conditioning pathway provides genuine geometric information and that our method successfully extracts a substantial fraction of the available signal without external data.

7 Ablation Studies

We isolate the contribution of each design decision through six ablation experiments.

7.1 Mechanism Identification: Layer Ablation

Pi3X’s conditioning interface operates through two pathways: (1) the *encode path*, which constructs ray embeddings and scale normalisation from conditioning poses and adds them to image features before decoding; and (2) the *decode path*, consisting of five PoseInjectBlocks that inject pose information via Projective Rotary Position Encoding (PRoPE) into decoder attention layers. To identify which pathway drives the self-refinement benefit, we ablate each independently.

Table 3: Layer ablation identifying the active conditioning pathway. Zeroing all five decode-path PoseInjectBlocks produces identical accuracy to leaving them active, while removing rotation or translation from the encode path causes catastrophic failure.

Configuration	ARE ($^\circ$)↓	Δ vs baseline
Full conditioning (all active)	3.85	−12%
Decode path zeroed (all 5 PoseInjectBlocks = 0)	3.84	−12%
Rotation-only conditioning ($\mathbf{t} = \mathbf{0}$)	116	catastrophic
Translation-only conditioning ($\mathbf{R} = \mathbf{I}$)	116	catastrophic
No conditioning (unconditioned baseline)	4.39	—

Key findings. (1) The decode-path PoseInjectBlocks contribute *nothing*: zeroing all five blocks yields $\text{ARE} = 3.84^\circ$, statistically identical to full conditioning (3.85°). The conditioning benefit resides entirely in the encode path. (2) Both rotation and translation are required for encode-path conditioning—either alone causes catastrophic failure (116°). This is because the projection matrix $P = \text{lift}(K) \cdot [R \mid \mathbf{t}]$ requires both components to construct valid ray embeddings. (3) Injection norms from PoseInjectBlocks decay across iterations ($5.90 \rightarrow 5.82 \rightarrow 5.79$), confirming that the blocks become progressively more dormant as the encode-path features improve—they are not contributing to the refinement, but responding to it.

Attention entropy. Cross-view attention entropy decreases with conditioning (higher certainty in feature matching), but this is a *consequence* of better encode-path features, not a cause of improved poses. The decode-path blocks that modulate attention are inert.

7.2 Effect of Hybrid Decoupling

Table 4 compares naïve iterative refinement (feeding the full conditioned output back without decoupling) against our hybrid approach.

Table 4: Naïve vs. hybrid iterative refinement. Naïve iteration improves rotation at the cost of translation; hybrid fusion achieves superior rotation improvement while preserving translation accuracy.

Method	ATE↓	ARE↓	Δ ATE	Δ ARE
Unconditioned baseline	0.161	4.39°	—	—
Naïve self-refine (iter 5)	0.195	3.60°	+21%	−18%
Hybrid MC (iter 2)	0.160	3.13°	−1%	−29%

Naïve iteration degrades ATE by 21% while improving ARE by only 18%. In contrast, hybrid fusion achieves 29% ARE improvement with negligible translation change. This confirms the theoretical prediction of Section 5: translation does not contract under self-conditioning and must be anchored to the unconditioned estimate.

7.3 Sensitivity to Conditioning Quality

To characterise the basin of attraction, we condition Pi3X with ground truth poses corrupted by varying levels of Gaussian noise (Table 5).

Table 5: Effect of conditioning pose quality on output accuracy. Translation noise is isotropic Gaussian in scene coordinates. Beyond $\sim 10^\circ$ rotation noise, conditioning becomes counterproductive and the model diverges.

Rot. noise (°)	Trans. noise	ATE↓	ARE↓
0 (perfect GT)	0	0.097	1.22°
1	0.05	0.087	1.30°
2	0.10	0.080	1.60°
5	0.20	0.077	3.34°
10	0.50	0.368	180°
20	1.00	0.951	180°

The model tolerates conditioning noise up to approximately 5° of rotation error, producing output that improves upon the conditioning input. At 10° , the conditioning signal becomes misleading and the model collapses to degenerate solutions (180° ARE indicates random orientations). The transition is sharp rather than gradual, indicating a well-defined basin boundary. Our self-conditioned prior ($\sim 4.4^\circ$ error) lies near the edge of this basin, which explains both why self-refinement succeeds and why it saturates after 2–3 iterations as the estimate approaches the basin’s contraction limit.

7.4 Convergence Behaviour

Table 6 reports per-iteration metrics for iterated hybrid fusion.

Table 6: Convergence of iterated hybrid fusion ($S = 1$, deterministic). Iteration 2 achieves the minimum ARE; iteration 4 produces an orientation flip in the rotation estimate.

Iteration	ATE	ARE ($^{\circ}$)	RPE _R ($^{\circ}$)	RPE _t
0 (unconditioned)	0.161	4.39	6.26	0.375
1	0.162	3.43	4.88	0.321
2	0.160	3.13	4.58	0.309
3	0.163	3.38	4.70	0.307
4	0.160	180 $^{\circ}$	4.78	8.55

The method converges monotonically through iterations 1–2, achieving a 29% ARE reduction. Iteration 3 shows slight regression, and iteration 4 produces catastrophic failure (orientation flip in at least one camera). This behaviour is consistent with the basin-of-attraction analysis: as the conditioned estimate approaches the $\sim 3^{\circ}$ fixed point, the margin between the estimate’s error and the basin boundary narrows, making the system sensitive to accumulated approximation error. We adopt a conservative stopping criterion of $M = 2$ iterations or ARE improvement below 0.1° between successive iterations.

7.5 Inefficacy of Sim(3) Re-alignment

One might hypothesise that translation degradation under naïve iteration is caused by scale drift—each iteration’s output uses a different arbitrary scale, and this accumulates. To test this, we Sim(3)-align each iteration’s output back to the conditioning frame before re-injection. Table 7 shows that this intervention has no measurable effect.

Table 7: Sim(3)-constrained vs. unconstrained naïve iteration. Constraining scale does not prevent translation degradation, confirming that the failure mechanism is systematic conditioning bias rather than scale drift.

Iter	Sim(3) ATE	Sim(3) ARE	Unconstrained ATE	Unconstrained ARE
0	0.152	4.69 $^{\circ}$	0.152	4.69 $^{\circ}$
1	0.160	3.50 $^{\circ}$	0.160	3.50 $^{\circ}$
2	0.180	3.31 $^{\circ}$	0.179	3.29 $^{\circ}$
5	0.197	3.61 $^{\circ}$	0.195	3.59 $^{\circ}$

The near-identical degradation trajectories confirm that translation failure is not a scale problem but a structural one: Pi3X’s conditioning mechanism introduces systematic translational bias that is independent of the output’s global scale. This motivates our decision to anchor translations entirely to the unconditioned source rather than attempting to correct the conditioned output.

7.6 Monte Carlo Perturbation

If the conditioning mechanism operated stochastically (e.g., via attention routing sensitive to pose perturbations), one might expect Monte Carlo integration—perturbing conditioning poses

and averaging outputs—to reduce error as $O(1/\sqrt{S})$ by sampling the model’s implicit posterior $p(R^* | R + \epsilon, \text{imgs})$.

We implement MC sampling with three design considerations that are critical for correctness in the Sim(3) setting. First, each perturbed sample produces poses in a different arbitrary Sim(3) frame; we apply Sim(3) Procrustes alignment across samples before aggregation. Second, rather than running the full pipeline independently per sample (which discards cross-sample information), we pool all $S \times K$ observations into a single enlarged dataset and run one pipeline pass on the full pool. Third, we use *structured perturbation*: a shared random rotation applied to all cameras simultaneously, preserving the inter-camera constellation geometry that Pi3X exploits, with only small per-camera residuals ($\sigma_R/5$) for diversity.

We evaluate this MC strategy with $M = 1$ iteration (to isolate the MC effect from iteration gains), sweeping perturbation magnitude $\sigma_R \in \{0.5^\circ, 1.0^\circ, 2.0^\circ\}$ and sample count $S \in \{2, 4, 8\}$. Table 8 reports the results.

Table 8: Monte Carlo posterior sampling ($M = 1$, structured perturbation, pool-before-pipeline with Sim(3) alignment). The $S = 1$ baseline (no perturbation) achieves $\text{ARE} = 3.63^\circ$. No configuration with $S > 1$ improves upon it.

σ_R ($^\circ$)	S	ARE ($^\circ$)↓	ATE↓	Cost
—	1	3.63	0.154	2×
0.5	2	3.64	0.153	3×
0.5	4	3.65	0.153	5×
0.5	8	3.63	0.153	9×
1.0	2	3.65	0.153	3×
1.0	4	3.65	0.153	5×
1.0	8	3.64	0.153	9×
2.0	2	3.67	0.153	3×
2.0	4	3.66	0.153	5×
2.0	8	3.63	0.153	9×

The results are unambiguous: no multi-sample configuration outperforms deterministic single-pass conditioning. The best cases ($S = 8$, $\sigma_R \in \{0.5^\circ, 2.0^\circ\}$) match $S = 1$ at 3.63° but never exceed it, despite $4.5\times$ additional compute. Larger perturbation magnitudes slightly degrade intermediate S values (e.g., $S = 2$ at $\sigma_R = 2^\circ$ yields 3.67°), indicating that the perturbation introduces error without eliciting useful posterior diversity.

This finding is consistent with the block coordinate descent interpretation (Section 5.3): since the conditioning mechanism operates through deterministic encode-path features (ray embeddings and scale normalisation), all perturbations within the basin of attraction produce the same enriched features and converge to the same output. The posterior is effectively a point mass (angular standard deviation $< 0.02^\circ$ across samples), leaving no variance for MC averaging to reduce. Deterministic iteration ($S = 1$, $M = 2$) remains strictly preferable, achieving $\text{ARE} = 3.19^\circ$ at $3\times$ cost.

8 Discussion

Practical deployment. The iterated hybrid fusion method ($M = 2$) provides the most favourable accuracy–cost tradeoff: two conditioned inference passes ($3\times$ total compute) yield 27% rotation improvement while preserving translation. The mechanism is deterministic feature enrichment via

the encode path—no stochastic sampling is needed, and Monte Carlo perturbation adds compute without benefit (Section 7.6). Hybrid decoupling is essential: naïve self-conditioning without it leads to progressive translation degradation.

Generality of the framework. While our experiments use Pi3X, the decoupled hybrid refinement framework applies in principle to any learned geometry model that (i) accepts pose conditioning as input, and (ii) outputs poses defined up to a gauge ambiguity. Critically, our ablation reveals that the conditioning must operate through the encode path (e.g., ray embeddings, scale normalisation) rather than solely through decode-path attention injection, which we show to be inert. The contraction analysis (Section 5) provides a diagnostic: measuring input–output error reduction under corrupted conditioning reveals whether a given model exhibits the contraction property and, if so, at what rate. Models with $\lambda < 1$ are amenable to self-refinement; those with $\lambda \geq 1$ are not.

Limitations. Several constraints bound the applicability of our method. The accuracy floor ($\sim 3^\circ$ ARE) is determined by the contraction rate and the model’s irreducible error—it cannot be lowered without stronger external priors or architectural modifications to the encode-path conditioning (e.g., breaking the Sim(3) scale ambiguity in ray construction). Translation accuracy is bounded by the quality of unconditioned inference, since hybrid fusion anchors translations to this source. The convergence is empirical: while iterations 1–3 consistently improve, iteration 4 and beyond can produce orientation flips (Table 6), necessitating a stopping criterion. Finally, our evaluation is conducted on a single multi-camera installation; generalisation to other wide-baseline deployments with different camera counts and geometries remains to be validated.

Future work. Several directions merit investigation. First, Pi3X also accepts depth conditioning—monocular depth priors (e.g., Depth Anything v2) could provide complementary geometric signal without requiring ground truth poses, potentially breaking the rotation floor. Second, partial conditioning (conditioning only high-confidence cameras while leaving others unconditioned) may exploit the model’s cross-view attention to propagate reliable pose information to uncertain views. Third, sparse geometric anchors (a pre-calibrated camera pair or LiDAR registration points) could provide a stronger initialisation that places the estimate deeper within the contraction basin, accelerating convergence and lowering the fixed point. Fourth, the tight unimodality of the posterior under self-conditioning suggests that the model has learned a near-deterministic mapping; investigating whether alternative perturbation strategies (e.g., perturbing image inputs rather than poses, or using dropout at inference) can elicit meaningful posterior diversity remains an open question.

9 Conclusion

We have shown that self-conditioning in learned geometry models exhibits a fundamental asymmetry: rotation contracts toward a fixed point on $SO(3)$ while translation diverges due to Sim(3) scale ambiguity. Exploiting this asymmetry via decoupled hybrid fusion—sourcing rotation from conditioned inference and translation from unconditioned inference—yields consistent improvements without the degradation of naïve iteration. Iterating twice achieves 27% rotation improvement over the raw Pi3X baseline on a challenging wide-baseline, feature-adversarial calibration task, at only $3\times$ the compute cost and with no external calibration data required.

Mechanistically, we trace the conditioning benefit to the encode path—ray embeddings and scale normalisation derived from approximate poses—rather than the decode-path PoseInjectBlocks, which we show contribute nothing when ablated. This identifies self-refinement as block coordinate

descent on a joint pose-feature objective: approximate poses yield ray direction priors that reduce feature ambiguity, which in turn yields better poses. The contraction rate ($\lambda \approx 0.8$) reflects the composite Lipschitz constant of this encode-decode loop restricted to $\text{SO}(3)$, and the resulting accuracy floor ($\sim 3^\circ$) is a property of the model and scene, not the method—breaking this floor requires either stronger external priors or architectural changes to the encode-path conditioning.

We believe the decoupled self-refinement framework generalises to any learned geometry model whose conditioning operates through encode-path feature enrichment, and the contraction analysis provides a principled diagnostic for predicting when iterative refinement will help.

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